28. Note that if $a < 0$ and $b < 0$, then $-a \in P$ and $-b \in P$ so $ab = (-a)(-b) \in P$, so $0 < ab$. It follows at once that a product of an even number of elements x_i where every $x_i < 0$ is an element of P. Thus any product of an even number of elements, all of which are greater than zero or all of which are less than zero, is sure to be positive. From $a^{2n+1} = b^{2n+1}$, we obtain $(a^2)^n a = (b^2)^n b$ and we see the either $a < 0$ and $b < 0$, or $0 < a$ and $0 < b$. Consider the factorization

$$0 = a^{2n+1} - b^{2n+1} = (a-b)(a^{2n} + a^{2n-1}b + a^{2n-2}b^2 + \cdots + b^{2n}).$$

Every summand in parentheses is the product of an even number of factors that are either a or b. Because a and b are either both greater than zero or both less than zero, every summand in parentheses is positive, and thus their sum is positive, and hence nonzero. Because R has no zero divisors, we must have $a - b = 0$, so $a = b$.

29. In the chart following, the order in the left column indicates the order in which the indeterminants were adjoined to R, and whether the order when they were adjoined was P_{high} or P_{low}. The other columns indicate whether the inequality the top of the column is true (T) or false (F). Because no two rows have the same sequence of T's and F's, the orderings are all different.

Ordering	$x < y$	$1 < x$	$1 < y$	$xy < 1$
x-high, y-high	T	T	T	F
x-high, y-low	F	T	F	T
x-low, y-high	T	F	T	F
x-low, y-low	F	F	F	T
y-high, x-high	F	T	T	F
y-high, x-low	T	F	T	T
y-low, x-high	F	T	F	F
y-low, x-low	T	F	F	T

26. Homomorphisms and Factor Rings

1. Let ϕ be a homomorphism of $\mathbb{Z} \times \mathbb{Z}$ into $\mathbb{Z} \times \mathbb{Z}$. Suppose that $\phi(1,0) = (m,n)$. From $\phi(1,0) = \phi[(1,0)(1,0)]$, we see that $m^2 = m$ and $n^2 = n$, so $\phi(1,0)$ must be one of the elements (0, 0), (1, 0), (0, 1), or (1, 1). By a similar argument , $\phi(0,1)$ must be one of these same four elements. We also must have $\phi(1,0)\phi(0,1) = \phi(0,0) = (0,0)$. This gives just 9 possibilities.

$\phi(1,0) = (1,0)$ while $\phi(0,1) = (0,0)$ or (0, 1),
$\phi(1,0) = (0,1)$ while $\phi(0,1) = (0, 0)$ or (1, 0),
$\phi(1,0) = (1, 1)$ while $\phi(0,1) = (0, 0)$, and
$\phi(1,0) = (0,0)$ while $\phi(0,1) = (0, 0)$, (1, 0), (0, 1) or (1, 1).

It is easily checked that each of these does give rise to a homomorphism.

2. In order for $\mathbb{Z}_n$ to contain a subring isomorphic to $\mathbb{Z}_2$, we see that Z_n must contain a nonzero element s such that $s + s = 0$ and $s^2 = s$, so that s can play the role of 1 in $\mathbb{Z}_2$. From $s + s = 0$, we see that n must be even. Let $n = 2m$, so that the group $\langle \{0, m\}, +_n \rangle \simeq \langle \mathbb{Z}_2, +_2 \rangle$. In order to have $\langle \{0, m\}, \cdot_n \rangle \simeq \langle \mathbb{Z}_2, \cdot_2 \rangle$, we must have $mm = m$. In $\mathbb{Z}_n$, we have $2 \cdot m = 0, 3 \cdot m = (2 \cdot m) + m = m, 4 \cdot m = 0, 5 \cdot m = m$, etc. Thus we have $mm = m$ in $\mathbb{Z}_n$ if and only if m is an odd integer. Hence $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$ if and only if $n = 2m$ for an odd integer m.

3. Because the ideals must be additive subgroups, by group theory we see that the possibilities are restricted to the cyclic additive subgroups

$$\begin{aligned}
&\langle 0\rangle = \{0\},\\
&\langle 1\rangle = \{0,1,2,3,4,5,6,7,8,9,0,10,11\},\\
&\langle 2\rangle = \{0,2,4,6,8,10\},\\
&\langle 3\rangle = \{0,3,6,9\},\\
&\langle 4\rangle = \{0,4,8\}, \text{ and}\\
&\langle 6\rangle = \{0,6\}.
\end{aligned}$$

It is easily checked that each of these is closed under multiplication by any element of $\mathbb{Z}_{12}$, so they are ideals. We have $\mathbb{Z}_{12}/\langle 0\rangle \simeq \mathbb{Z}_{12}$, $\mathbb{Z}_{12}/\langle 1\rangle \simeq \{0\}$, $\mathbb{Z}_{12}/\langle 2\rangle \simeq \mathbb{Z}_2$, $\mathbb{Z}_{12}/\langle 3\rangle \simeq \mathbb{Z}_3$, $\mathbb{Z}_{12}/\langle 4\rangle \simeq \mathbb{Z}_4$, and $\mathbb{Z}_{12}/\langle 6\rangle \simeq \mathbb{Z}_6$.

4. Here are the tables for addition and multiplication in $2\mathbb{Z}/8\mathbb{Z}$.

$+$	$8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$
$8\mathbb{Z}$	$8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$
$2+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$8\mathbb{Z}$
$4+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$8\mathbb{Z}$	$2+8\mathbb{Z}$
$6+8\mathbb{Z}$	$6+8\mathbb{Z}$	$8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$

$\cdot$	$8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$
$8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$
$2+8\mathbb{Z}$	$8\mathbb{Z}$	$4+8\mathbb{Z}$	$8\mathbb{Z}$	$4+8\mathbb{Z}$
$4+8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$	$8\mathbb{Z}$
$6+8\mathbb{Z}$	$8\mathbb{Z}$	$4+8\mathbb{Z}$	$8\mathbb{Z}$	$4+8\mathbb{Z}$

The rings $2\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}_4$ are not isomorphic, for $2\mathbb{Z}/8\mathbb{Z}$ has no unity while $\mathbb{Z}_4$ does.

5. The definition is incorrect; ϕ must map R onto R'.

An **isomorphism** of a ring R with a ring R' is a homomorphism $\phi : R \to R'$ mapping R onto R' such that $\text{Ker}(\phi) = \{0\}$.

6. The definition is correct.

7. The definition is incorrect. The set description is nonsense.

The **kernel of a homomorphism** ϕ mapping a ring R into a ring R' is $\{r \in R \mid \phi(r) = 0'\}$.

8. The differentiation map δ is not a homomorphism; $\delta(f(x)g(x)) = f'(x)g(x) + f(x)g'(x) \neq f'(x)g'(x) = \delta(f(x))\delta(g(x))$. To connect this with Example 26.12, we note that the kernel of δ as an additive group homomorphism is the set of all constant functions. Example 26.12 shows that this is not an ideal, so δ cannot be a ring homomorphism.

9. Let $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be defined by $\phi(n) = (n, 0)$. Then $\mathbb{Z}$ has unity 1, but $\phi(1) = (1, 0)$ is not the unity of $\mathbb{Z} \times \mathbb{Z}$; the unity of $\mathbb{Z} \times \mathbb{Z}$ is (1, 1).

10. T F T F T F T T T T

11. (See the text answer.)

12. We know that $\mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}_2$, which is a field.

13. $\mathbb{Z}$ is an integral domain. $\mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}_4$, where 2 is a divisor of 0.

14. $\mathbb{Z} \times \mathbb{Z}$ has divisors of zero, but $(\mathbb{Z} \times \mathbb{Z})/(\mathbb{Z} \times \{0\}) \simeq \mathbb{Z}$ which has no divisors of zero.

15. $\{(n,n) \mid n \in \mathbb{Z}\}$ is a subring of $\mathbb{Z} \times \mathbb{Z}$, but it is not an ideal because $(2,1)(n,n) = (2n,n)$.

16. **a.** The notations r and s would be used to denote elements of R, not of R/N. The student probably does not understand the structure of a factor ring.

b. Assume that R/N is commutative. Then

$$(r+N)(s+N) = (s+N)(r+N) \text{ for all } r,s \in R.$$

c. Let $r, s \in R$. Then

$(r+N)(s+N) = (s+N)(r+N)$ for all $r, s \in R$
if and only if $rs + N = sr + N$ for all $r, s \in R$, so
if and only if $(rs+N) - (sr+N) = N$ for all $r, s \in R$, so
if and only if $(rs - sr) + N = N$ for all $r, s \in R$, so
if and only $(rs - sr) \in N$ for all $r, s \in R$.

17. Because $(a+b\sqrt{2})+(c+d\sqrt{2}) = (a+c)+(b+d)\sqrt{2}$ and $0 = 0+0\sqrt{2}$ and $-(a+b\sqrt{2}) = (-a)+(-b)\sqrt{2}$, we see that R is closed under addition, has an additive identity, and contains additive inverses. Thus $\langle R, + \rangle$ is a group. Now $(a+b\sqrt{2})(c+d\sqrt{2}) = (ac+2bd)+(ad+bc)\sqrt{2}$, so R is closed under multiplication and is thus a ring. We will show that R' is a ring by showing that it is the image of R under a homomorphism $\phi : R \to M_2$. Let

$$\phi(a+b\sqrt{2}) = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}.$$

Then

$$\begin{aligned} \phi((a+b\sqrt{2})+(c+d\sqrt{2})) &= \phi((a+c)+(b+d)\sqrt{2} \\ &= \begin{bmatrix} a+c & 2(b+d) \\ b+d & a+c \end{bmatrix} = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix} + \begin{bmatrix} c & 2d \\ d & c \end{bmatrix} \\ &= \phi(a+b\sqrt{2}) + \phi(c+d\sqrt{2}) \end{aligned}$$

and

$$\begin{aligned} \phi((a+b\sqrt{2})(c+d\sqrt{2})) &= \phi((ac+2bd)+(ad+bc)\sqrt{2}) \\ &= \begin{bmatrix} ac+2bd & 2(ad+bc)) \\ ad+bc & ac+2bd \end{bmatrix} = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix} \begin{bmatrix} c & 2d \\ d & c \end{bmatrix} \\ &= \phi(a+b\sqrt{2})\phi(c+d\sqrt{2}). \end{aligned}$$

This shows that ϕ is a homomorphism. Because $R' = \phi[R]$, we see that R' is a ring. If $\phi(a+b\sqrt{2})$ is the matrix with all entries zero, then we must have $a = b = 0$, so $\text{Ker}(\phi) = 0$ and ϕ is one to one. Thus ϕ is an isomorphism of R onto R'.

18. Let $\phi : F \to R$ be a homomorphism of a field F into a ring R, and let $N = \text{Ker}(\phi)$. If $N \neq \{0\}$, then N contains a nonzero element u of F which is a unit. Because N is an ideal, we see that $u^{-1}u = 1$ is in N, and then N contains $a1 = a$ for all $a \in F$. Thus N is either $\{0\}$, in which case ϕ is one to one by group theory, or $N = F$, so that ϕ maps every element of F onto 0.

19. By Exercise 49 of Section 13, $\psi\phi(r+s) = \psi\phi(r) + \psi\phi(s)$ for all $r, s \in R$. For multiplication, we note that $\psi\phi(rs) = \psi(\phi(rs)) = \psi(\phi(r)\phi(s)) = [\psi\phi(r)][\psi\phi(s)]$ because both ϕ and ψ are homomorphisms. Thus $\psi\phi$ is also a homomorphism.

20. In a commutative ring R, the binomial expansion $(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}$ is valid. If p is a prime and $n = p$, then all the binomial coefficients $\binom{p}{i}$ for $1 \leq i \leq p-1$ are divisible by p, and thus the term $\binom{p}{i} a^i b^{p-i} = 0$ for a and b in a commutative ring of characteristic p. This shows at once that $\phi_p(a+b) = (a+b)^p = a^p + b^p = \phi_p(a) + \phi_p(b)$. Also $\phi_p(ab) = (ab)^p = a^p b^p$ because R is commutative. But $a^p b^p = \phi_p(a)\phi_p(b)$, so ϕ is a homomorphism.

21. By Theorem 26.3, we know $\phi(1)$ is unity for $\phi[R]$. Suppose that R' has unity $1'$. Then $\phi(1) = \phi(1)1' = \phi(1)\phi(1)$ so that we have $\phi(1)1' - \phi(1)\phi(1) = 0'$. Consequently, $\phi(1)(1' - \phi(1)) = 0'$. Now if $\phi(1) = 0'$, then $\phi(a) = \phi(1a) = \phi(1)\phi(a) = 0'\phi(a) = 0'$ for all $a \in R$, so $\phi[R] = \{0'\}$ contrary to hypothesis. Thus $\phi(1) \neq 0'$. Because R' has no divisors of zero, we conclude from $\phi(1)(1' - \phi(1)) = 0'$ that $1' - \phi(1) = 0'$, so $\phi(1)$ is the unity $1'$ of R'.

22. a. Because the ideal N is also a subring of R, Theorem 26.3 shows that $\phi[N]$ is a subring of R'. To show that is is an ideal of $\phi[R]$, we show that $\phi(r)\phi[N] \subseteq \phi[N]$ and $\phi[N]\phi(r) \subseteq \phi[N]$ for all $r \in R$. Let $r \in R$ and let $s \in N$. Then $rs \in N$ and $sr \in N$ because N is an ideal. Applying ϕ, we see that $\phi(r)\phi(s) = \phi(rs) \in \phi[N]$ and $\phi(s)\phi(r) = \phi(sr) \in \phi[N]$.

b. Let $\phi : \mathbb{Z} \to \mathbb{Q}$ be the injection map given by $\phi(n) = n$ for all $n \in \mathbb{Z}$. Now $2\mathbb{Z}$ is an ideal of $\mathbb{Z}$, but $2\mathbb{Z}$ is not an ideal of $\mathbb{Q}$ because $(1/2)2 = 1$ and 1 is not in $2\mathbb{Z}$.

c. Let N' be an ideal of R' or of $\phi[R]$. We know that $\phi^{-1}[N']$ is at least a subring of R by Theorem 26.3. We must show that $r\phi^{-1}[N'] \subseteq \phi^{-1}[N']$ and that $\phi^{-1}[N']r \subseteq \phi^{-1}[N']$ for all $r \in R$. Let $s \in \phi^{-1}[N']$, so that $\phi(s) \in N'$. Then $\phi(rs) = \phi(r)\phi(s)$ and $\phi(r)\phi(s) \in N'$ because N' is an ideal. This shows that $rs \in \phi^{-1}[N']$, so $r\phi^{-1}[N'] \subseteq \phi^{-1}[N']$. Also $\phi(sr) = \phi(s)\phi(r)$ and $\phi(s)\phi(r) \in N'$ because N' is an ideal. This shows that $sr \in \phi^{-1}[N']$, so $\phi^{-1}[N']r \subseteq \phi^{-1}[N']$.

23. If $f(x_1, \cdots, x_n)$ and $g(x_1, \cdots, x_n)$ both have every element of S as a zero, then so do their sum, product, and any multiple of one of them by any element $h(x_1, \cdots, x_n)$ in $F[x_1, \cdots, x_n]$. Because the possible multipliers from $F[x_1, \cdots, x_n]$ include 0 and -1, we see that the set N_S is indeed a subring closed under multiplication by elements of $F[x_1, \cdots, x_n]$, and thus is an ideal of this polynomial ring.

24. Let N be an ideal of a field F. If N contains a nonzero element a, then N contains $(1/a)a = 1$, because N is an ideal. But then N contains $s1 = s$ for every $s \in F$, so $N = F$. Thus N is either $\{0\}$ or F. If $N = F$, then $F/N = F/F$ is the trivial ring of one element. If $N = \{0\}$, then $F/N = F/\{0\}$ is isomorphic to F, because each element $s + \{0\}$ of $F/\{0\}$ can be renamed s.

25. If $N \neq R$, then the unity 1 of R is not an element of N, for if $1 \in N$, then so is $r1 = r$ for all $r \in R$. Thus $1 + N \neq N$, that is, $1 + N$ is not the zero element of R/N. Clearly $(1+N)(r+N) = r + N = (r+N)(1+N)$ in R/N, which shows that $1 + N$ is unity for R/N.

26. Let $x, y \in I_a$ so $ax = ay = 0$. Then $a(x+y) = ax + ay = 0 + 0 = 0$ so $(x+y) \in I_a$. Also, $a(xy) = (ax)y = 0y = 0$ so $xy \in I_a$. Because $a0 = 0$ and $a(-x) = -(ax) = -0 = 0$, we see that I_a contains 0 and additive inverses of each of its elements x, so I_a is a subring of R. (Note that thus far, we have not used commutativity in R.) Let $r \in R$. Then $a(xr) = (ax)r = 0r = 0$ so $xr \in I_a$, and because R is commutative, we see that $a(rx) = r(ax) = r0 = 0$, so $rx \in I_a$. Thus I_a is an ideal of R.

27. Let $\{N_i \mid i \in I\}$ be a collection of ideals in R. Each of these ideals is a subring of R, and Exercise 49 of Section 18 shows that $N = \bigcap_{i \in I} N_i$ is also a subring of R. We need only show that N is closed under multiplication by elements of R. Let $r \in R$ and let $s \in N$. Then $s \in N_i$ for all $i \in I$. Because each N_i is an ideal of R, we see that $rs \in N_i$ and $sr \in N_i$ for all $i \in I$. Thus $rs \in N$.

28. By Exercise 39 of Section 14, the map $\phi_* : R/N \to R'/N'$ defined by $\phi_*(r+N) = \phi(r) + N'$ is well defined and satisfies the additive requirements for a homomorphism. Now we have $\phi_*((r+N)(s+N)) = \phi_*(rs+N) = \phi(rs)+N' = [\phi(r)\phi(s)]+N' = [\phi(r)+N'][\phi(s)+N'] = [\phi_*(r+N][\phi_*(s+N]$ so ϕ_* also satisfies the multiplicative condition, and is a ring homomorphism.

29. The condition that ϕ maps R *onto* a *nonzero* ring R' shows that no unit of R is in $\mathrm{Ker}(\phi)$, for if $\mathrm{Ker}(\phi)$ contains a unit u, then it contains $(ru^{-1})u = r$ for all $r \in R$, which would mean that $\mathrm{Ker}(\phi) = R$ and R' would be the zero ring.

Let u be a unit in R. Because $\phi[R] = R'$, we know that $\phi(1)$ is unity $1'$ in R'. From $uu^{-1} = u^{-1}u = 1$, we obtain $\phi(uu^{-1}) = \phi(u)\phi(u^{-1}) = 1'$ and $\phi(u^{-1}u) = \phi(u^{-1})\phi(u) = 1'$. Thus $\phi(u)$ is a unit of R', and its inverse is $\phi(u^{-1})$.

30. Let $\sqrt{\{0\}}$ be the collection of all nilpotent elements of R. Let $a, b \in \sqrt{\{0\}}$. Then there exist positive integers m and n such that $a^m = b^n = 0$. In a *commutative* ring, the binomial expansion is valid. Consider $(a+b)^{m+n}$. In the binomial expansion, each summand contains a term $a^i b^{m+n-i}$. Now either $i \geq m$ so that $a^i = 0$ or $m+n-i \geq n$ so that $b^{m+n-i} = 0$. Thus each summand of $(a+b)^{m+n}$ is zero, so $(a+b)^{m+n} = 0$ and $\sqrt{\{0\}}$ is closed under addition. For multiplication, we note that because R is commutative, $(ab)^{mn} = (a^m)^n(b^n)^m = (0)(0) = 0$, so $ab \in \sqrt{\{0\}}$. If $s \in R$, then $(sa)^m = a^m s^m = 0s^m = 0$ so $\sqrt{\{0\}}$ is also closed under left and right multiplication by elements of R. Taking $x = 0$, we see that $0 \in \sqrt{\{0\}}$. Also $(-a)^m$ is either a^m or $-a^m$, so $(-a)^m = 0$ and $-a \in \sqrt{\{0\}}$. Thus $\sqrt{\{0\}}$ is an ideal of R.

31. The nilradical of $\mathbb{Z}_{12}$ is $\{0, 6\}$. The nilradical of $\mathbb{Z}$ is $\{0\}$ and the nilradical of $\mathbb{Z}_{32}$ is $\{0, 2, 4, 6, 8, \cdots, 30\}$.

32. Suppose $(a+N)^m = N$ in R/N. Then $a^m \in N$. Because N is the nilradical of R, there exists $n \in \mathbb{Z}^+$ such that $(a^m)^n = 0$. But then $a^{mn} = 0$ so $a \in N$. Thus $a + N = N$ so $\{N\}$ is the nilradical of R/N.

33. Let $a \in R$. Because the nilradical of R/N is R/N, there is some positive integer m such that $(a+N)^m = N$. Then $a^m \in N$. Because every element of N is nilpotent, there exists a positive integer n such that $(a^m)^n = 0$ in R. But then $a^{mn} = 0$, so a is an element of the nilradical of R. Thus the nilradical of R is R.

34. Let $a, b \in \sqrt{N}$. Then $a^m \in N$ and $b^n \in N$ for some positive integers m and n. Precisely as in the answer to Exercise 30, we argue that $(a+b) \in N$, $ab \in N$, and also that $sa \in N$ and $as \in N$ for any $s \in R$. Because $0^1 \in N$, we see that $0 \in \sqrt{N}$. Also $(-a)^m$ is either a^m or $-(a^m)$, and both a^m and $-(a^m)$ are in N. Thus $-a \in \sqrt{N}$. This shows that $\sqrt{N}$ is an ideal of R.

35. **a.** Let $R = \mathbb{Z}$ and let $N = 4\mathbb{Z}$. Then $\sqrt{N} = 2\mathbb{Z} \neq 4\mathbb{Z}$.

b. Let $R = \mathbb{Z}$ and let $N = 2\mathbb{Z}$. Then $\sqrt{N} = 2\mathbb{Z}$.

36. If $\sqrt{N}/N$ is viewed as a subring of R/N, then it is the nilradical of R/N, in the sense of the definition in Exercise 30.

37. We have

$$\begin{aligned} \phi[(a+bi)+(c+di)] &= \phi[(a+c)+(d+b)i] = \begin{bmatrix} a+c & b+d \\ -b-d & a+c \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} c & d \\ -d & c \end{bmatrix} = \phi(a+bi) + \phi(c+di). \end{aligned}$$